Chapter 3 Quiz — Energy and Speed Are Related

Name: _____ Date: _____

Read each question carefully. Do your best!

1.

Speed is _____.

a) a measurement of the distance an object travels over an amount of time

b) how heavy an object is c) a kind of energy d) a visible substance

2.

A moving object has _____.

a) stored energy only b) energy of motion c) no energy d) only sound energy

3.

Which comparison gives the BEST evidence about speed and energy?

a) a small ball vs. a big ball, both moving slowly

b) the SAME ball moving slowly, then the SAME ball moving faster

c) a fast car vs. a slow bicycle d) a marble vs. a bowling ball

4. READ THE DATA

A group rolled ONE ball down a ramp. Here is their data:

Ramp Height	Average Cup Displacement
1 book	5 cm
2 books	12 cm
3 books	20 cm

At which height did the ball cause the greatest change? _____

Did the group measure the ball's speed, or the change it caused? Explain.

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5. READ THE GRAPH

A graph titled “Ramp Height vs. Average Cup Displacement” shows the bar gets taller as ramp height increases.

What does this graph suggest about speed and energy of motion for the SAME ball?

6.

Write M (measured) or I (inferred) next to each item.

____ the cup moved 18 cm

____ the ball was moving faster

____ the faster ball had more energy of motion

____ the ramp height was 3 books

7.

The SAME ball is thrown gently, then thrown hard. The hard throw makes your hand sting.
Use evidence to explain: which throw had greater energy of motion, and how do you know?

8.

A student says, “We measured the ball's energy with our ruler.” What is wrong with this statement?

9.

For the SAME object, faster motion is associated with _____ energy of motion.

a) less b) the same c) greater d) no

10. (BONUS)

Why is it important to keep the object the SAME when comparing speed and energy?
